3. TJEDAN

~ Snaga i energija Fourierovog reda i Fourierove transformacije ~

Zadatak 1.) Izračunajte energiju sljedećih kontinuiranih signala

$$\underbrace{E = \int_{-\infty}^{\infty} |f(t)|^2 \, dt}_{\text{ovo nas traži}} \qquad P = \lim_{L\to\infty} \frac{1}{L} \int_{-\frac{L}{2}}^{\frac{L}{2}} |f(t)|^2 \, dt \quad \text{za CT}$$

a) $x(t) = e^{-at}\mu(t)$, $a>0$ — vremenski kontinuirani signal [CT]

$$E = \int_{-\infty}^{\infty} e^{-2at} \cdot \mu^2(t) = \int_0^{\infty} e^{-2at} \, dt = \frac{-1}{2a} e^{-2at}\Big|_0^{\infty} = \frac{-1}{2a}\left(\underbrace{\frac{1}{\infty}}_{0} - \frac{1}{1}\right) = \boxed{\frac{1}{2a}}$$

b) $x(t) = t\mu(t) \longrightarrow E = \int_{-\infty}^{\infty} (t\mu(t))^2 \, dt = \int_0^{\infty} t^2 \, dt = \frac{1}{3} t^3 \Big|_0^{\infty} = \boxed{\infty}$

Zadatak 2.) Nađite energiju sljedećih disk. signala

$$\underbrace{E = \sum_{n=-\infty}^{\infty} |f[n]|^2}_{} \qquad P = \lim_{M\to\infty} \frac{1}{2M+1} \sum_{n=-M}^{M} |f[n]|^2$$

a) $x[n] = (-0.5)^n \mu[n]$ — vremenski diskretni signal [DT]

$$E = \sum_{n=-\infty}^{\infty} \left|(-0.5)^{2n} \mu^2[n]\right| = \sum_{n=0}^{\infty} (-0.5)^{2n} = \sum_{n=0}^{\infty} \left(\frac{-1}{2}\right)^{2n} = \sum_{n=0}^{\infty} \left(\frac{1}{4}\right)^n = \text{geom.}$$

$$E = \frac{1}{1-\frac{1}{4}} \Rightarrow \boxed{E = \frac{4}{3}}$$

b) $x[n] = n(\mu[n] - \mu[n-5]) = n\mu[n] - n\mu[n-5]$

$$E = \sum_{n=-\infty}^{\infty} |x[n]|^2 = \sum_{n=-\infty}^{\infty} n^2\mu[n] - \sum_{n=-\infty}^{\infty} n^2\mu[n-5] = \sum_{n=0}^{\infty} n^2 - \sum_{n=5}^{\infty} n^2 = \sum_{n=0}^{4} n^2 \Rightarrow \boxed{30}$$

Zadatak 3.) Zadan je diskretan signal $x[n] = n\left(\mu[n] - \mu[n-2008]\right)$. Izračunajte energiju signala.

$$\sum_{n=-\infty}^{\infty} |n\mu[n]|^2 - \sum_{n=-\infty}^{\infty} |n\mu[n-2008]|^2 = \sum_{n=0}^{\infty} n^2 - \sum_{n=2008}^{\infty} n^2 = \sum_{n=0}^{2007} n^2 \Rightarrow \boxed{E = 2691679140}$$

Zadatak 4.) Izračunajte srednju snagu kontinuiranog signala

$$x(t) = e^{j2\pi t} \cdot \sin\left(t + \frac{\pi}{3}\right)$$

$$P = \lim_{L\to\infty} \frac{1}{L} \int_{-\frac{L}{2}}^{\frac{L}{2}} |x(t)|^2 = \lim_{L\to\infty} \int_{-\frac{L}{2}}^{\frac{L}{2}} \left|e^{j2\pi t} \cdot \sin\left(t + \frac{\pi}{3}\right)\right|^2$$

$$e^{j2\pi t} = \cos(2\pi t) + j\sin(2\pi t)$$

$$P = \lim_{L\to\infty} \frac{1}{L} \int_{-\frac{L}{2}}^{\frac{L}{2}} |e^{j2\pi t}|^2 \cdot \left|\sin\left(t + \frac{\pi}{3}\right)\right|^2 dt$$

$$|e^{j2\pi t}| = \sqrt{\underbrace{\cos^2(2\pi t) + \sin^2(2\pi t)}_{1}} = 1$$

$$P = \lim_{L\to\infty} \frac{1}{L} \int_{-\frac{L}{2}}^{\frac{L}{2}} \sin^2\left(t + \frac{\pi}{3}\right) dt$$

$$\cos 2x = \cos^2 x - \sin^2 x$$
$$= 1 - \sin^2 x - \sin^2 x$$
$$= 1 - 2\sin^2 x$$
$$\rightarrow \sin^2 x = \frac{1 - \cos 2x}{2}$$

$$P = \lim_{L\to\infty} \frac{1}{L} \cdot \frac{1}{2} \int_{-\frac{L}{2}}^{\frac{L}{2}} 1 - \cos 2\left(t + \frac{\pi}{3}\right) dt$$

$$P = \lim_{L\to\infty} \frac{1}{2L} \int_{-\frac{L}{2}}^{\frac{L}{2}} dt - \lim_{L\to\infty} \frac{1}{2L} \int_{-\frac{L}{2}}^{\frac{L}{2}} \cos 2\left(t + \frac{\pi}{3}\right) dt$$

$$P = \lim_{L\to\infty} \frac{1}{2\cancel{L}} \cancel{\left(\frac{L}{2} + \frac{L}{2}\right)} - \lim_{L\to\infty} \frac{1}{2L} \int_{-\frac{L}{2}}^{\frac{L}{2}} \cos 2\left(t + \frac{\pi}{3}\right) dt$$

$$P = \frac{1}{2} - \frac{1}{2} \lim_{L\to\infty} \frac{1}{L} \cdot \frac{1}{2} \sin\left(t + \frac{\pi}{3}\right)\Big|_{-\frac{L}{2}}^{\frac{L}{2}} = \frac{1}{2} - \frac{1}{4} \lim_{L\to\infty} \frac{1}{L} \left(\sin\left(\frac{L}{2} + \frac{\pi}{3}\right) - \sin\left(\frac{-L}{2} + \frac{\pi}{3}\right)\right)$$

$$P = \frac{1}{2} - \frac{1}{4} \lim_{L\to\infty} \frac{1}{L} \cdot 2\sin \frac{\frac{L}{2} + \cancel{\frac{\pi}{3}} + \frac{L}{2} - \cancel{\frac{\pi}{3}}}{2} \cdot \cos \frac{\cancel{\frac{L}{2}} + \frac{\pi}{3} - \cancel{\frac{L}{2}} + \frac{\pi}{3}}{2} =$$

$$P = \frac{1}{2} - \frac{1}{2} \lim_{L\to\infty} \frac{1}{L} \sin\frac{L}{2} \cdot \underbrace{\cos\frac{\pi}{3}}_{\frac{1}{2}} = \frac{1}{2} - \cancel{\frac{1}{4} \underbrace{\lim_{L\to\infty} \frac{1}{L} \cdot \sin\frac{L}{2}}_{0}} \rightarrow \boxed{P = \frac{1}{2}}$$

Zadatak 5.) Nađite srednju snagu diskretnih signala

a) $x[n] = \mu[n]$

$$P = \lim_{M\to\infty} \frac{1}{2M+1} \sum_{n=-M}^{M} |x[n]|^2 = \lim_{M\to\infty} \frac{1}{2M+1} \sum_{n=0}^{M} 1$$

$$\to P = \lim_{M\to\infty} \frac{1}{2M+1} \cdot (M+1) = \lim_{M\to\infty} \frac{M+1}{2M+1} = \lim_{M\to\infty} \frac{1+\frac{1}{M}}{2+\frac{1}{M}} = \boxed{\frac{1}{2}}$$

b) $x[n] = 2e^{j3n}$

$$P = \lim_{M\to\infty} \frac{1}{2M+1} \sum_{n=-M}^{M} 4|e^{j3n}|^2 = \lim_{M\to\infty} \frac{4}{2M+1} \sum_{n=-M}^{M} |e^{j3n}|^2$$

$$|e^{j3n}| = \sqrt{\cos(3n)^2 + \sin(3n)^2} \Rightarrow 1 \qquad P = \lim_{M\to\infty} \frac{4}{2M+1} \underbrace{\sum_{n=-M}^{M} 1}_{2(M+1)}$$

(jer ide od $n=-M$)

$$P = \lim_{M\to\infty} \frac{4}{\cancel{2M+1}} \cdot \cancel{(2M+1)} \Rightarrow \boxed{P = 4}$$

Zadatak 6.) Izračunajte kut između sljedećih signala $f_k(t) = \cos\left(\frac{2\pi}{T} k t\right)$ za $k \in \mathbb{Z}$

→ definiran na domeni $[0, T]$.

$$\cos\varphi = \frac{\langle f | g \rangle}{\|f\|_2 \cdot \|g\|_2} \qquad \langle f | g \rangle = \int_{-\infty}^{\infty} f(t) g^*(t)\, dt$$

$$\|f\|_2 = \sqrt{\int_{-\infty}^{\infty} |f(t)|^2 dt} \neq \|f_1\| = \int_{-\infty}^{\infty} (f(t))^2 dt$$

$f_1 = \cos\left(\frac{2\pi}{T} n t\right)$, $f_2 = \cos\left(\frac{2\pi}{T} m t\right)$ → za sve $m, n \in \mathbb{Z}$, $m \neq n$

$$\langle f_1 | f_2 \rangle = \int_{-\infty}^{\infty} \cos\left(\frac{2\pi}{T} n t\right) \cdot \cos\left(\frac{2\pi}{T} m t\right) dt = \int_{-\infty}^{\infty} \frac{1}{2}\left[\cos\left(\frac{2\pi}{T} t (m-n)\right) + \cos\left(\frac{2\pi}{T} t (m+n)\right)\right] dt$$

$$= \frac{1}{2}\left[\int_{-\infty}^{\infty} \cos\left(\frac{2\pi}{T}(m-n) t\right) dt + \int_{-\infty}^{\infty} \cos\left(\frac{2\pi}{T}(m+n) t\right) dt\right]$$

$$= \frac{1}{2}\left[\frac{T}{2\pi(m-n)} \sin\left(\frac{2\pi}{T}(m-n) t\right)\Big|_{-\infty}^{\infty} + \frac{T}{2\pi(m+n)} \sin\left(\frac{2\pi}{T}(m+n) t\right)\Big|_{-\infty}^{\infty}\right]$$

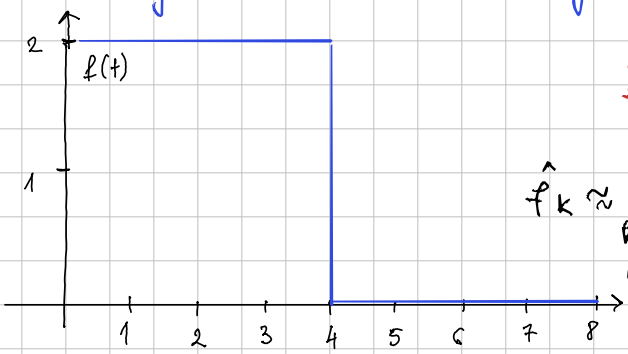
$$= \frac{T}{4\pi}\left[\frac{1}{m-n}\left(\underbrace{\sin(2\pi(m-n))}_{0} - \underbrace{\sin(0)}_{0}\right) + \frac{1}{m+n}\left(\underbrace{\sin(2\pi(m+n))}_{0} - \underbrace{\sin(0)}_{0}\right)\right]$$

$$\to \langle f_1 | f_2 \rangle = 0 \Rightarrow \boxed{\cos\varphi = 0}$$ međusobno okomiti

Zadatak 7.) Odrediti najbolju aproksimaciju $f(t): [0,8] \to \mathbb{R}$ zadanog slikom, pomoću signala $\hat{f}_k(t)$ koji je linearna kombinacija osnovnih signala $\psi_0(t) = a$, $t \in [0,8]$, gdje je $a>0$ konstanta, i $\psi_k(t) = \sin(\frac{\pi}{4}kt)$ za $t \in [0,8]$ i $k \in \mathbb{Z} \setminus \{0\}$.

Odredite vrijednost konstante a tako da svi osnovni signali $\psi_k(t)$ imaju jednaku energiju. Odredite koef. rastlaganja F_k za $k=-1,0,1$.

Osnovni signali su međusobno ortogonalni.



$$\hat{f}_k = \sum_{k=-K}^{K} \frac{\langle f, \psi_k \rangle}{\|\psi_k\|_2^2} \cdot \psi_k$$

$$\hat{f}_k \approx \sum_{k=-K}^{K} F_k \cdot \psi_k = f(t)$$

$$\cos 2x = 1 - 2\sin^2 x$$
$$\sin^2 x = \frac{1-\cos 2x}{2}$$

$$\|\psi_0\|_2^2 = \int_0^8 a^2\,dt = t a^2 \Big|_0^8 = 8a^2$$

$$\|\psi_k\|_2^2 = \int_0^8 \sin^2\left(\frac{\pi}{4}kt\right) dt = \frac{1}{2}\int_0^8 1 - \cos\left(\frac{\pi}{2}kt\right) dt = 4 - \frac{1}{2}\int_0^8 \cos\left(\frac{\pi}{2}kt\right) dt$$

$$= 4 - \frac{1}{\cancel{2}} \cdot \frac{\cancel{2}}{k\pi} \cdot \sin\left(\frac{\pi}{2}kt\right)\Big|_0^8 = 4 - \frac{1}{k\pi}\left(\underbrace{\sin(4\pi k)}_{0} - \underbrace{\sin(0)}_{0}\right) \Rightarrow \boxed{\|\psi_k\|_2 = 4}$$

$$\boxed{\|\psi_0\|_2^2 = \|\psi_k\|_2^2} \Rightarrow 8a^2 = 4 \to a^2 = \frac{1}{2}$$

da bi svi osnovni signali imali jednaku energiju

$$a = \frac{1}{\sqrt{2}}$$

Koeficijenti rastlaganja: $F_k = \dfrac{\langle f | \psi_k \rangle}{\|\psi_k\|_2^2}$

$f(t) \to$ na intervalu $[0,8]$ funkcija $f(t) = 2$ na $[0,4]$

$$F_0 = \frac{\langle f | \psi_0 \rangle}{\|\psi_0\|_2^2} = \frac{1}{4}\langle f | \psi_0 \rangle = \frac{1}{4} \cdot \int_0^8 f(t) \cdot \psi_0\,dt = \frac{1}{\cancel{4}_{2}} \int_0^4 2 \cdot \frac{1}{\sqrt{2}}\,dt$$

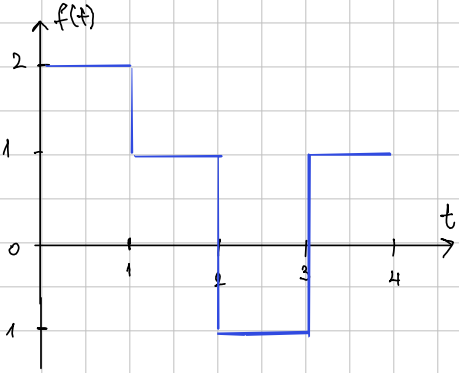
$$\boxed{F_0 = \frac{1}{2\sqrt{2}} \int_0^4 dt = \frac{2}{\sqrt{2}}}$$

$$F_k = \frac{1}{4}\int_0^8 \sin\left(\frac{\pi}{4}kt\right) f(t)\,dt = \frac{1}{4}\int_0^4 2 \cdot \sin\left(\frac{\pi}{4}kt\right) dt = \frac{1}{\cancel{4}} \cdot 2 \cdot \frac{4}{\pi k} \cos\left(\frac{\pi}{4}kt\right)\Big|_0^4$$

$$= \frac{2}{\pi k}\left(\cos(\pi k) - \underbrace{\cos(0)}_{1}\right) \longrightarrow \boxed{F_k = \frac{2}{\pi k}\left(\cos(\pi k) - 1\right)}$$

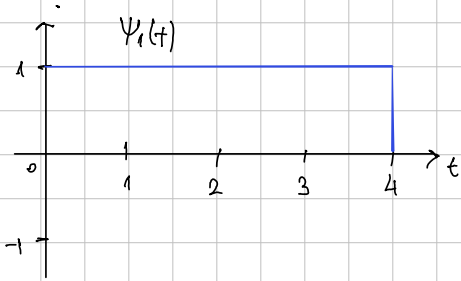
$$\boxed{F_1 = \frac{-4}{\pi}} \qquad \boxed{F_{-1} = \frac{4}{\pi}}$$

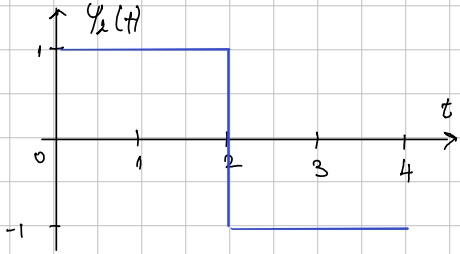
Zadatak 8.) $f(t) : [0,4] \longrightarrow \mathbb{R}$

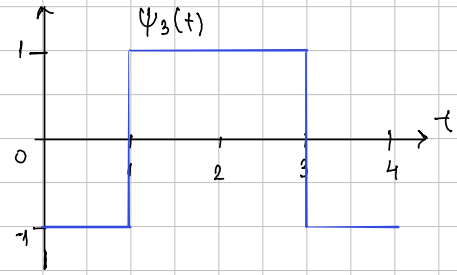


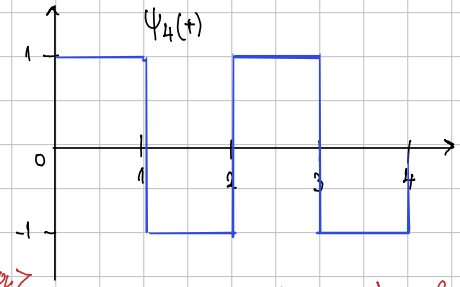
Rastavite zadani signal po osnovnim signalima $\psi_1(t)$, $\psi_2(t)$, $\psi_3(t)$, $\psi_4(t)$ prikazanim na slici ispod.

Zadani osnovni signali su međusobno ortogonalni i jednakih energija.









traži nas koeficijent $F_k = \dfrac{\langle f|\psi_k\rangle}{\|\psi_k\|_2^2}$

budući da su jednakih energija dovoljno je izračunati $\|\psi\|_2^2$ za bilokoji od ψ.

isto i ovaj $\langle f|\psi\rangle$

$$\|\psi_k\|_2^2 \rightarrow \|\psi_1\|_2^2 = \sqrt{\int_0^4 |\psi_1|^2 dt}^{\,2} = \int_0^4 1\,dt = 4$$

$$\langle f|\psi_1\rangle = \int_{-\infty}^{\infty} f(t)\cdot\psi(t)\,dt = \int_0^1 2\cdot 1\,dt + \int_1^2 1\cdot 1\,dt + \int_2^3 (-1)\cdot 1\,dt + \int_3^4 1\cdot 1\,dt$$

$$\langle f|\psi_1\rangle = 3$$

$$\boxed{F_k = \frac{\langle f|\psi_1\rangle}{\|\psi_1\|_2^2} = \frac{3}{4}}$$

Zadatak 9.) Zadan je diskretni signal $x[n] = \{\underline{2}, -2, 0, 1\}$, $x \in \ell_2[0,3]$. Rastavite zadani signal po osnovnim signalima

Zadanim sa:

$$\psi_1[n] = \frac{1}{2} \times \{\underline{1}, 1, 1, 1\}, \quad \psi_1 \in \ell_2[0,3]$$

$$\psi_2[n] = \frac{1}{2} \times \{\underline{1}, 1, -1, -1\}, \quad \psi_2 \in \ell_2[0,3]$$

$$\psi_3[n] = \frac{1}{\sqrt{2}} \times \{\underline{1}, -1, 0, 0\}, \quad \psi_3 \in \ell_2[0,3]$$

$$\psi_4[n] = \frac{1}{\sqrt{2}} \times \{\underline{0}, 0, 1, -1\}, \quad \psi_4 \in \ell_2[0,3]$$

Zadani osnovni signali su međusobno ortogonalni i jednakih energija.

$$F_k = ? \qquad F_k = \frac{\langle f | \psi \rangle}{\|\psi\|_2^2} = \frac{\langle f | \psi \rangle}{\ell_2}$$

→ budući da se radi o diskretnim signalima ⟹ radimo sa sumom umjesto integralima

$$\|\psi\|_2^2 = \sqrt{\sum_{-\infty}^{\infty} |\psi|^2}^{\,2} = \sqrt{\sum_{n=0}^{3} |\psi_1|^2}^{\,2} = \sum_{n=0}^{3} \left(\frac{1}{2}\right)^2 = \left(\frac{1}{2}\right)^2 \cdot 4 = \underline{\underline{1}}$$

$$\langle f | g \rangle = \sum_{n=-\infty}^{\infty} f[n]^* g[n] \rightarrow \langle x | \psi_1 \rangle = \sum_{n=0}^{3} x[n]^* \psi_1[n]$$

$$\langle x | \psi_1 \rangle = 2 \cdot \cancel{\frac{1}{2}} + \cancel{(-2) \cdot \frac{1}{2}} + 0 \cdot \cancel{\frac{1}{2}} + 1 \cdot \frac{1}{2} = \underline{\underline{\frac{1}{2}}} \qquad \boxed{F_1 = \frac{1}{2}}$$

$$\langle x | \psi_2 \rangle = 2 \cdot \cancel{\frac{1}{2}} + \cancel{(-2) \cdot \frac{1}{2}} + 0 \cdot \cancel{\frac{1}{2}(-1)} + 1 \cdot \frac{1}{2}(-1) = \frac{-1}{2} \qquad \boxed{F_2 = \frac{-1}{2}}$$

$$\langle x | \psi_3 \rangle = 2 \cdot \frac{1}{\sqrt{2}} + (-2) \cdot \left(\frac{-1}{\sqrt{2}}\right) + 0 + 1 \cdot 0 = \frac{4}{\sqrt{2}} \qquad \boxed{F_3 = \frac{4}{\sqrt{2}}}$$

$$\langle x | \psi_4 \rangle = \cancel{2 \cdot 0 + (-2) \cdot 0} + 0 \cdot \cancel{1 \cdot \frac{1}{\sqrt{2}}} + 1 \cdot \frac{1}{\sqrt{2}} \cdot (-1) \rightarrow \boxed{F_4 = \frac{-1}{\sqrt{2}}}$$